

The J-space in OLMo-3-32B-Think: an open-weights replication of “Verbalizable Representations Form a Global Workspace in Language Models”

special-lab-1, interpretability labs

July 26, 2026

run 2026-07-25_1726 · lens j1ens @ 581d3986 · seeds 0

v2 delta run in progress (§10): results below marked [v2] update as instruments land

Abstract

We replicate the core claims of Anthropic’s July 2026 “global workspace” paper on **Olmo-3-32B-Think**, a fully open (weights + training data) reasoning model, using the reference Jacobian-lens implementation. The *geometry* replicates: a sparse, verbalizable subspace concentrated in the middle layers, whose directions are read by 73–94 downstream components (vs. 0 for random directions). Its *capacity* is roughly an order of magnitude thinner than reported for Claude ($\leq 0.67\%$ of activation variance vs. 6–10%; ~ 6 active concepts vs. ~ 25). The paper’s headline *causal dissociation does not replicate* at our doses: static J-space ablation is indistinguishable from random controls, while matched-dimension *non-verbalizable* high-variance directions cause the real selective damage. Exploiting the thinking variant, we add a result the paper could not test: the workspace does *not* contain the answer before reasoning begins (median rank 5613), but **leads the chain of thought by a median 46 tokens** during reasoning, and is loaded on demand when thinking is suppressed (rank 211 at the answer position, collapsing monotonically L30→L60). An eval-awareness probe finds a highly consistent “being-evaluated” readout direction ($p < 10^{-3}$ on all probed layers) that is semantically legible yet behaviorally inert under ablation.

1 Setup

Model. allenai/Olmo-3-32B-Think (64 layers, $d = 5120$), bf16 on one 96 GB RTX PRO 6000 Blackwell — full-precision tier: no quantization, no gradient checkpointing. **Lens.** Reference j1ens package (anthropics/jacobian-lens), fit on 120 WikiText-103 prompts $\times$ 128 tokens (the neuronpedia reference recipe; j1ens README reports saturation near 100; the paper used 1000). Source layers: dense middle band L20–44 (step 2) + early/late controls {4,8,12,16,48,52,56,60}; target L63; four 30-prompt slices merged via `JacobianLens.merge`. Fit cost 5.2 h (~ 157 s/prompt, peak 80.6 GB); per-prompt max $\|J\|/\sqrt{d}$ stable at 0.94–0.95 and running-mean deltas fell $\sim 1/n$ (converged). **Dictionary.** Concept i at layer ℓ read through the RMSNorm-gain linearization ($W_U \odot g$) J_ℓ , rows normalized. **Sanity gate.** Easy single-token factual probes: 17/21 hit rank ≤ 20 mid-band for *both* J-lens and logit lens (ceiling). The discriminative check, multihop bridge-entity readout at the final prompt token ($n = 60$): J-lens pass@1 **0.283** vs. logit-lens 0.200 (+41% rel.); pass@20 0.547 vs. 0.586 — J-lens sharpens the rank-1 readout without expanding the top-20 shortlist, the same “replicates, less clean” flavor as Nanda’s Qwen 3.6 27B replication.

2 How to read the instruments (reader’s model)

Three orientation points carry every result below; the full version with worked examples is `report/REPORT_v2.md` §“How to read this lab”.

(1) **The lens is a fitted, averaged Jacobian.** $J_\ell = \partial \text{logits} / \partial h(\ell)$ — all downstream layers collapsed into one linear map — estimated *once* over the fitting corpus and frozen; readout is a matrix multiply, no per-prompt backprop. Dictionary row v is a *push* direction (“moving $h(\ell)$ this way most raises token v ’s final logit”), not output-space resemblance — which is why a concept can be J-visible while logit-lens-invisible if its coordinates only become output-aligned many layers later, and why the J-advantage lives in the mid band. One linearization for all inputs is also the method’s noise source (and why the fitting corpus is a real degree of freedom). (2) **The**

readout claim is not the workspace claim. Finding the hidden bridge — item 0: “the language spoken in the country where the Amazon River ends” → Brazil (never stated) → *Portuguese* — makes a better logit lens. The workspace claim is a *conjunction*: small + localized + broadcast + **causally privileged**. This lab tests the conjunction; it breaks at the causal joint. **(3) Read causal cells with the prior inverted.** Each ablation removes <1% of dimensions carrying a measured ≈ 0.5 –1.1% of activation energy — the default expectation is *nothing happens*, and the energy-matched random control delivers exactly that. In database terms: *static span* drops one small shared temp-table (same directions for every prompt) — only the dissociation pattern “multi-join queries fail, single-table lookups survive, fluency survives” would count as a workspace, and that pattern never appears on either model; *frozen per-item* deletes the row (rank the dictionary on the item’s prompt, freeze top-10) — its headline is expected, its value is the conjunction (a linear projection sufficed; the random-dictionary twin does nothing; *amnesia not aphasia* — generations stay coherent; and the 1-hop/2-hop damage shape classifies content-vs- scratchpad); *live per-token* re-selects each step and so deletes each word as it is born — maximally dramatic, minimally informative (the community replication’s confound, here measured with a matched-live control). If “deleting key ideas breaks output” feels unsurprising: correct — that instinct, formalized with controls, is the analysis. The paper needed the *unexpected* pattern; both models delivered the expected one, so the earned label is *content channel* (with the P5 rescue and Qwen’s spared one-hop as the two epilogues that keep it interesting).

3 Descriptive geometry: the workspace exists, ten times thinner

Sparse nonnegative gradient-pursuit decomposition of 200 diverse prompts (factual QA, chained arithmetic, code/SQL, prose) onto the dictionary:

- **Sparsity.** Median active concepts per (position, layer): **6** at $\theta=0.01$, 3 at $\theta=0.02$, $k_{90} \approx 4$, peaking at L36–42. Paper: ~ 10 –25.
- **Variance share.** $\leq 0.67\%$ of centered activation variance inside the band (0.91% at L60, unembedding-adjacent). Paper: 6–10%. The capacity gap ($\sim 10\times$) is threshold-robust.
- **Localization.** The cleanest signature is top-1 persistence across adjacent positions: 0.19–0.20 at L32–40 vs. 0.03–0.08 at both ends — an inverted U at 50–66% depth (paper band: 33–92%).
- **Broadcast — with a catch.** Top J -directions at L24/32/40 are read (input-projection alignment $z > 3$) by **73–94** downstream components vs. **0** for matched random directions ($p \approx 10^{-24}$). But matched-dimension top *non- J* principal components are read by 77–125: wide readout is a property of high-variance directions generally, not of verbalizable ones specifically.

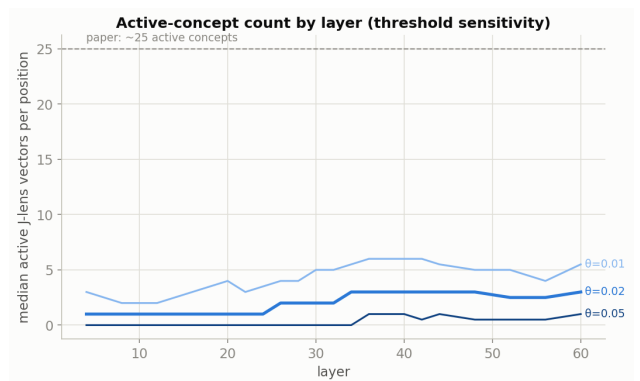
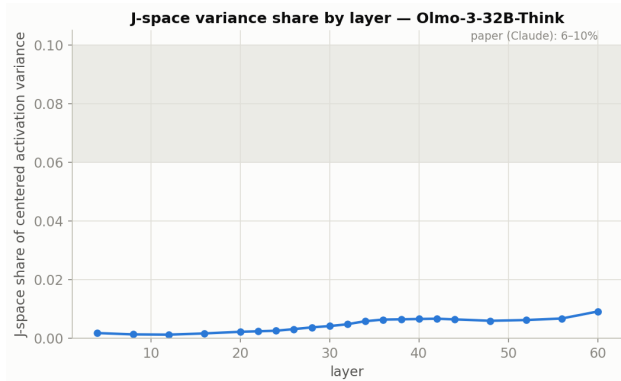


Figure 1: **Left:** J-space share of activation variance by layer; shaded band = paper’s Claude range (6–10%). **Right:** median active-concept count by layer under three activation thresholds; dashed line = paper’s ~ 25 .

4 Causal ablation: the dissociation does not replicate

Battery: 2-hop probe-swap facts ($n=60$), chained arithmetic ($n=30$), 3-table SQL with join-key checks ($n=30$), vs. one-hop facts, grammatical minimal pairs, held-out prose NLL. Conditions (band L20–44, doses $k \in \{10, 20, 40\}$ dims/layer, 2000-resample bootstrap CIs): static top- k J-span; matched random subspace; matched top non-J PCs; paper-faithful per-token dynamic top-10.

condition	2-hop	2-hop Δ lp	arith	SQL	1-hop	grammar	NLL
none (baseline)	0.58	—	0.87	0.67	0.60	1.00	2.71
J-space dyn top-10	0.00	−23.0	0.00	0.00	0.00	0.30	24.34
J-span $k=20$	0.52	−0.17	0.93	0.67	0.60	1.00	2.85
random $k=20$	0.52	−0.08	0.90	0.67	0.60	1.00	2.75
non-J PCA $k=20$	0.28	−0.86	1.00	0.00	0.70	1.00	3.45

Table 1: Accuracy (and 2-hop answer-logprob delta vs. baseline) under ablation; $k=10/40$ follow the same pattern. Static J-span $\approx$ random everywhere; the *non-verbalizable* control causes the selective damage.

Three observations. (1) The paper-faithful *dynamic* condition removes whatever the model is currently computing — output degenerates to asterisks, NLL 2.71→24.34 — reproducing the “live-computation confound” Nanda flagged; it cannot count as evidence for a workspace. (2) Static J-span ablation is statistically indistinguishable from random subspaces on every task *and* on the finer answer-logprob instrument (Δ −0.15... −0.18 vs. −0.01... −0.11; CIs straddle 0). (3) The non-J high-variance control produces the only structured damage — and generation samples show its signature is *register destruction*: SQL completions lose the formal register entirely (SELECT the user’s city and the user’s country, and the user’s name, ...) while arithmetic content survives. At these doses, OLMo’s causal load sits in generic high-variance directions, not in the verbalizable subspace.

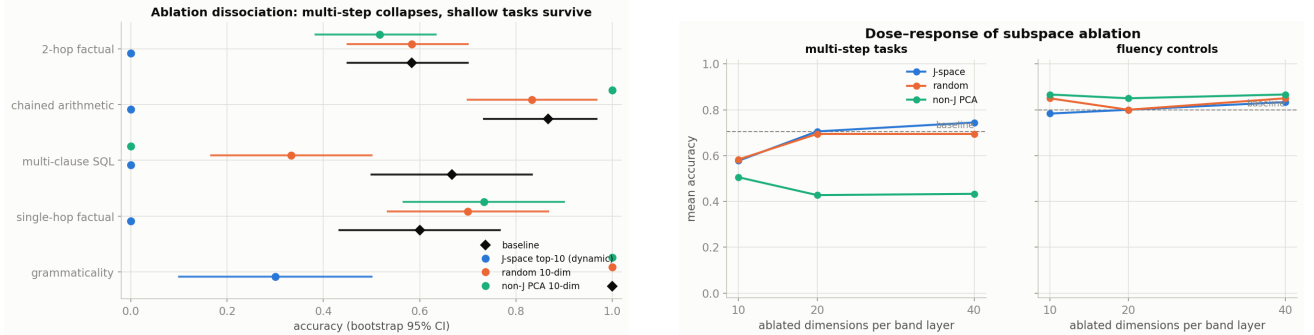


Figure 2: **Left:** per-task accuracy with 95% bootstrap CIs under the $k=10$ conditions. **Right:** dose-response for multi-step vs. fluency task groups.

5 The thinking-model angle: the workspace leads the chain of thought

Olmo-3-32B-Think emits explicit reasoning tokens, so the paper’s identity claim (workspace content = verbalizable reasoning substrate) becomes testable in time. 90 items (40 two-hop, 30 chained arithmetic, 20 SQL join-key); J-readout at L24/32/40 at every generation step; suppressed-CoT variant closes the template’s open <think> immediately.

- **Pre-CoT anticipation: null.** At the final prompt token the answer’s median J-rank is **5613** (10% of items ≤ 20); logit lens 11610 (0%). The workspace does not precompute the answer before reasoning.
- **Intra-CoT anticipation: strong.** The answer enters a read layer’s top-8 during thinking in 82–100% of items, and on traced items ($n=34$) the workspace holds the answer a **median 46 steps before the CoT first states it** (2-hop 48.5, SQL 38, arithmetic 9.5; workspace first in 91% of items).

- **Answer-time loading.** Forcing an immediate answer (closed `</think>`, zero reasoning tokens) pulls the answer’s median rank from 5613 to **211** ($23\% \leq 20$), collapsing monotonically L30→L60: 2048 → 1549 → 519 → 305 → 118 → 63 → 10 → 6, with J-lens ahead of logit lens through L28–40. Suppression costs accuracy: 2-hop 0.95 → 0.82, arithmetic 1.00 → 0.90, SQL 1.00 → 1.00.
- **Faithfulness probe.** 116 events where a stable workspace top-1 word never appears in the nearby CoT (59/90 items) — but only 4 involve the final answer. The recurring divergent words are *discourse-mode* adverbs: **famously** (stable top-1 for 7–10 steps while the CoT recalls “Springfield ... birthplace of basketball”), **respectively**, **etc** during enumerations. We find no “workspace holds X, CoT argues Y, answer is X” case; the workspace appears to track the discourse mode of the current reasoning span rather than concealed content.

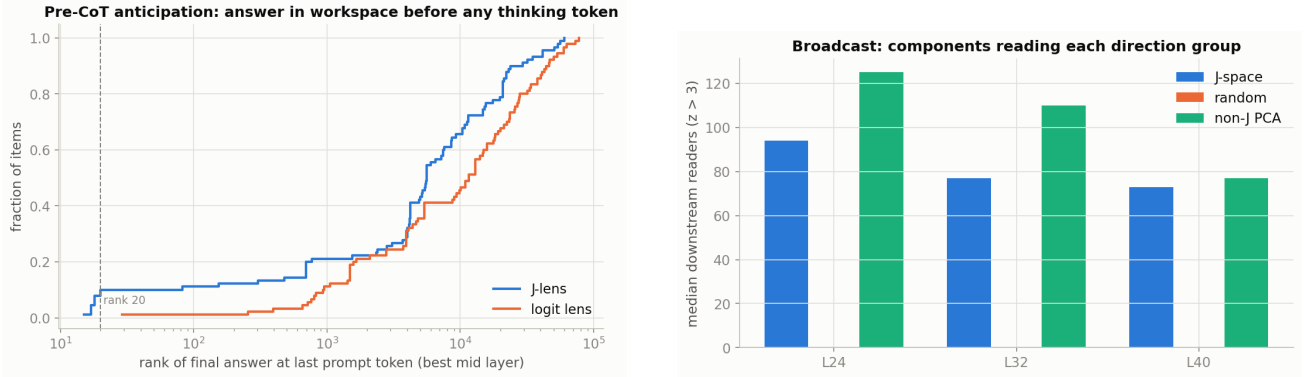


Figure 3: **Left:** ECDF of the answer’s rank at the last prompt token (best mid layer), J-lens vs. logit lens — anticipation is absent in both. **Right:** median downstream readers per direction group (broadcast).

6 Stretch: an eval-awareness direction, legible but inert

Twenty tasks rendered neutrally vs. prefixed with an explicit “formal evaluation, you are being tested and scored” framing; paired J-readout differencing at the final prompt token, seven band layers. A consistent direction emerges far beyond the pre-registered criterion: sign-flip permutation $p < 10^{-3}$ at *every* layer, mean pairwise cosine of paired deltas 0.48–0.71, mean-delta norm growing with depth (22→134). From L32 upward its top decoded tokens are literally the evaluation lexicon (**evaluation**, **assessed**, **assessing**, **eval**); at L24–28 the associates are obliquer (**monitored**, **internally**, **careful**). Yet the gated mini-ablation — projecting out the mean Δh direction during generation — changed essentially nothing. Because the framing states the cue words outright, this result is best read as a clean demonstration that the J-space *verbalizes context*, not as evidence of a latent “I am being tested” state; the confound-breaking variant (implied, unstated cues) is future work.

7 Side-by-side

8 Limitations

(1) Dictionary uses the RMSNorm-gain linearization ($W_U \odot g$) J_ℓ ; the final block’s full nonlinearity is not inverted. (2) Fitted band 20–44 (33–70% depth) $\subset$ the paper’s 33–92%; the suppressed-CoT profile (rank still collapsing L44→60) suggests unfit late-band structure. (3) Active-concept thresholds are not perfectly commensurable with the paper’s pursuit coefficients; the variance-share gap is the firmer capacity statement. (4) The non-J control is top-variance, not variance-*matched*; its larger effect partly reflects more energy removed — the single most important follow-up. (5) The dynamic ablation deletes live computation (Nanda’s confound); a frozen prompt-selected top- k is the clean middle instrument, not yet run. (6) WikiText corpus (not Dolma); 120 fit prompts; single seed; modest n (30–60/task); CoT tasks near ceiling compress suppression contrasts. (7) Phase-5 framing states its cue words (semantic echo); its ablation was inert.

	Claude (paper)	Qwen 3.6 27B (Nanda)	Olmo-3-32B-Think (ours)
Lens recipe	1000 prompts	25 Pile prompts	120 WikiText prompts, 21 layers
Active concepts	~10–25	not headline	6 ($\theta=0.01$); 3 ($\theta=0.02$)
Variance share	6–10%	—	$\leq 0.67\%$ ($\sim 10\times$ thinner)
Localization	33–92% depth	bands “less clean”	inverted U, peak 50–66% depth
Broadcast	wide readout	—	73–94 vs. 0 random; but non-J PCs 77–125
Causal dissociation	top-10 ablation kills multi-hop, spares shallow	flagged as possibly confounded	null : static J $\approx$ random; non-J PCs do the damage; dynamic = confounded collapse
Workspace vs. CoT	identity claim; CoT partially rescues	poetry/arith evals failed	pre-CoT null; leads CoT by 46 steps ; answer-time loading under suppression

Table 2: Replication scorecard. Paper/Nanda entries as pinned in the lab’s PLAN/LOG from the 2026-07-25 fetches.

9 What to run next

(1) Variance-matched non-J controls for broadcast and ablation — energy vs. content. (2) Frozen prompt-selected top-10 ablation during generation. (3) CoT-rescue: static ablations under think-mode prompting (the paper predicts externalization rescues; our suppression asymmetry predicts it here). (4) Fit L46–62 and redo the readouts where answer formation actually happens. (5) Dolma-corpus lens — the scientific-legibility payoff of doing this on OLMO. (6) **famously**-attractor forensics across models. (7) Implied-cue eval-awareness with a behavioral endpoint.

10 The v2 delta run: resolving the open instruments

The v1 verdict left a live question: *is the causal null a property of OLMO or of our instruments?* The published Qwen comparison is narrower than “Qwen has a J-space” — its readout evals replicated while its causal interventions were flagged by the replicator himself as possibly confounded — and v1 has three known instrument gaps, each capable of producing a false null: the non-J control was not energy-matched (its damage could be an energy artifact), the static J-span is position-agnostic (could miss position-specific workspace structure), and the fitted band stops at 70% depth while v1’s own suppressed-CoT profile shows answer formation continuing to L60. The v2 run closes these in order (variance matching, frozen prompt-selected ablation, late-band fit), calibrates the cot-lead headline against a false-positive floor, then tests CoT rescue. Decision tree: if the clean instruments flip the causal result, v1’s null was harness; if they confirm it, the same instruments go to Qwen 3.6 27B (Neuronpedia’s published lens, our harness) to separate model from method.

10.1 [v2-P4] The cot-lead headline survives its false-positive floor

The v1 claim — workspace holds the answer a median 46 steps before the CoT states it — used a permissive detector (any case/space variant first-token in any of 3 read layers’ top-8, across hundreds of steps), so any word gets many chances to fire. Running the *identical* detector on matched foils (fig. 4): frequency-matched random content words fire at **0.06** (vs. **0.92** for answers) — the pure noise floor is $15\times$ below the signal. Same-family wrong answers (other bridge entities, other join columns, nearby numbers) fire at 0.32 — but where they also appear in the CoT text their median lead is **3 steps** (vs. 46): family foils show up in the workspace *concurrently* with the text discussing them, exactly what a faithful context-verbalizing readout should do while the model enumerates candidates. Within items, the answer is detected earlier than *every* one of its foils in 66% of items (median answer detection rank 1). The 46-step lead is specific to the answer, not an artifact of detector permissiveness.

10.2 [v2-P1a] v1’s controls were energy-mismatched by an order of magnitude — measured, then fixed

Streaming the raw residual stream over the 200-prompt descriptive set and measuring exactly what each ablation hook removes: the k-dim J-span carries 0.5–1.8% of raw activation energy ($k=10/20/40$), which equals **25–33 / 44–54 / 80–94 random dimensions** — or just **1–2 / 2–5 / 3–8 deep non-J PCs**. Both of v1’s rank-matched

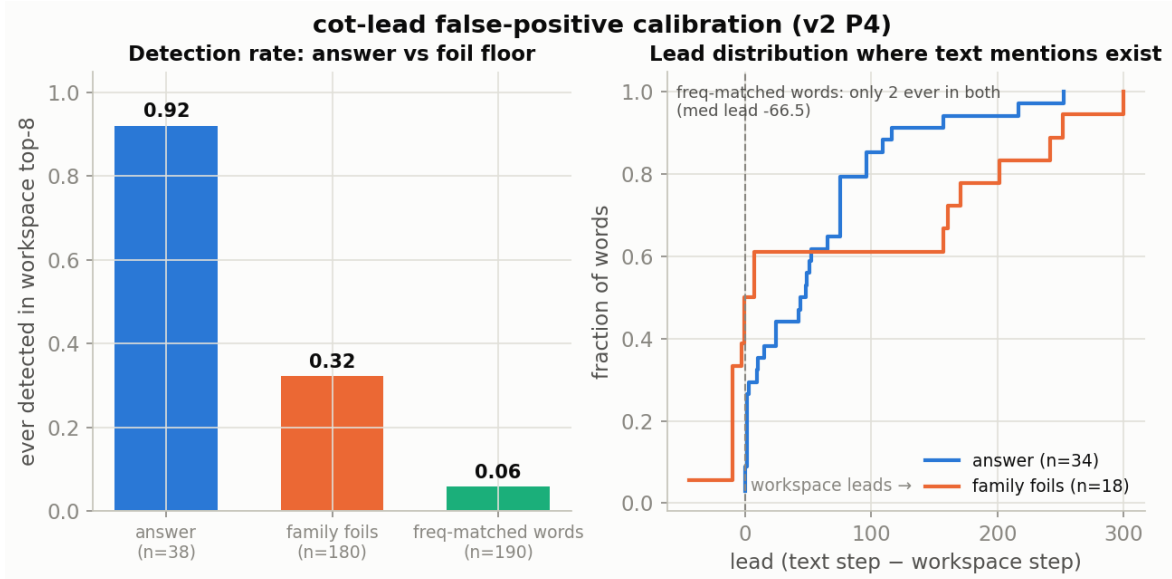


Figure 4: Foil calibration of the cot-lead detector. **Left:** detection rates — answers 0.92, family foils 0.32, frequency-matched words 0.06. **Right:** lead distributions where text mentions exist; the long positive lead is answer-specific.

comparisons were therefore energy-confounded, in opposite directions: **random_k** removed $\sim 3\times$ *less* energy than the J-span (which makes v1’s “J $\approx$ random” *conservative* — J removed more energy and still did no more damage), while **nonJ_pca_k** (top-k PCs) removed far *more* (its damage is suspect). The v2 matched controls hit the J-span’s removed energy to within 1% at every band layer and dose (fig. 5); the top PCs are so energy-fat that a contiguous *window* of deeper PCs (e.g. PCs 55–60) was needed at all 13 layers. The matched causal grid is running; its result decides whether v1’s “non-J does the real damage” was content or energy.

10.3 [v2-P1b] At matched energy the causal null is clean: damage tracks energy, not content

The energy-matched grid (fig. 6) settles the question P1a raised. Every piece of v1’s apparent non-J selectivity disappears: SQL returns to baseline at every dose (0.667 vs. v1’s unmatched 0.000 — the “register destruction” was pure energy), the two-hop answer-logprob deltas shrink from $-0.45/ -0.86/ -0.93$ (CIs excluding zero) to $-0.01/ -0.17/ -0.15$ (every CI straddling zero), and prose NLL sits at 2.74–2.79 against a 2.71 baseline (v1 unmatched: up to 4.02). With the confound removed in both directions, the three groups are statistically indistinguishable from each other *and from baseline* at every dose up to 40 dims/layer:

2-hop Δ logprob vs. baseline	k=10	k=20	k=40
J-span	-0.15	-0.17	-0.18
random (matched)	-0.11	-0.09	-0.08
non-J PCs (matched)	-0.01	-0.17	-0.15

Two consequences. First, the v1 causal verdict is *confirmed and strengthened*: on OLMo, statically removing the verbalizable subspace does nothing that removing an energy-equivalent arbitrary subspace doesn’t do — there is no content-specific causal load at these doses, and the earlier appearance of one was an instrument artifact we have now measured and eliminated. Second, v1’s own “J $\approx$ random” comparison was *conservative*: the J-span carries $2\text{--}5\times$ more energy than rank-matched random dims, so v1 was comparing a heavier J-removal against a lighter control and still finding parity. The last static escape hatch for the paper’s dissociation on OLMo is position-specific structure — frozen prompt-selected ablation (P2, running) and the late band (P3).

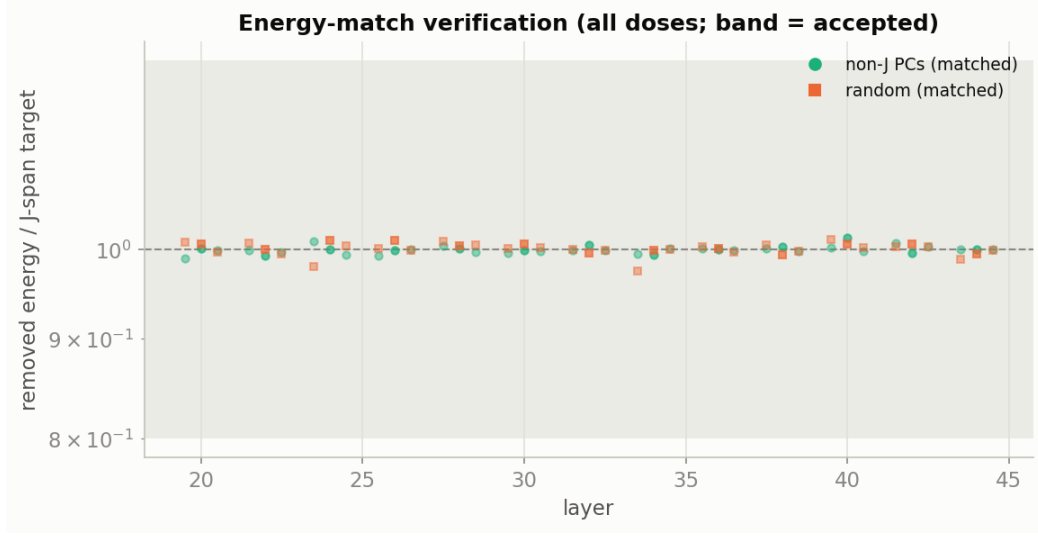


Figure 5: Energy-match verification: removed energy of each matched control relative to its J-span target (log scale), all band layers and doses. Every point sits on the 1.0 line, inside the pre-registered acceptance band.

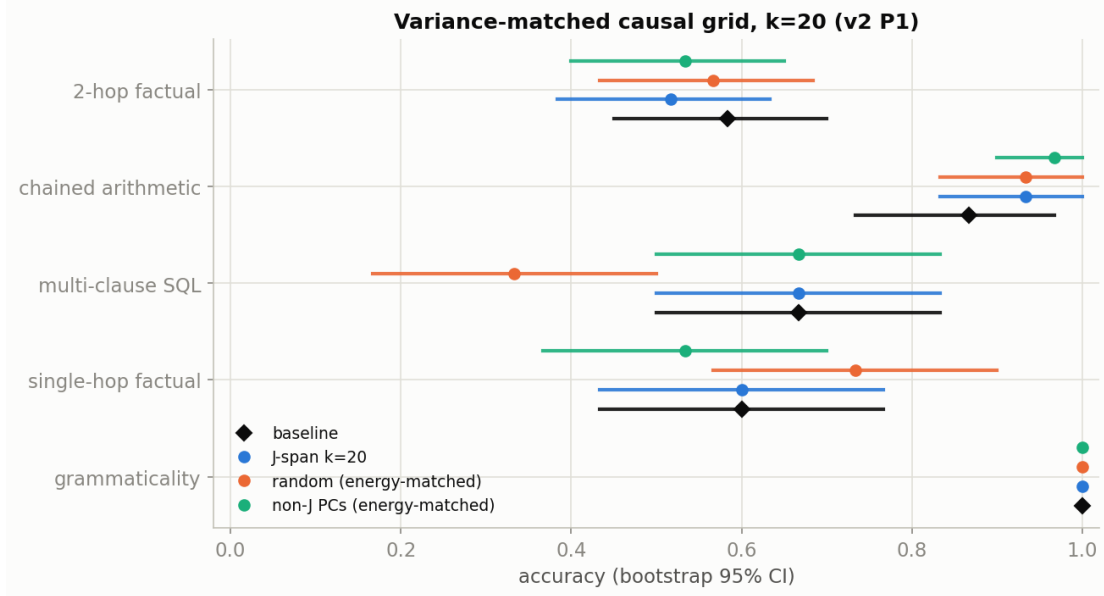


Figure 6: Energy-matched causal grid at k=20 (all doses in `v2_ablation_v2.json`). All three subspace families sit on top of each other and on baseline; contrast the v1 grid in §3, where the unmatched non-J control cratered SQL and two-hop.

10.4 [v2-P2] The frozen instrument finds the first content-specific causal handle — and it is recall, not reasoning

The confound-free variant of the paper’s intervention: per item, one clean forward pass ranks the J-dictionary by activity over the prompt, the top-10 per band layer are *frozen*, and generation proceeds with that static per-item projector. Two controls: the same mechanism on a random dictionary (**frozen_rand10**), and per-token live selection from a random dictionary (**live_rand10**, the matched-live control v1’s dynamic condition never had).

	2-hop	2-hop lp	1-hop	arith	SQL	NLL	grammar
baseline	0.58	−1.74	0.60	0.87	0.67	2.71	1.00
frozen J-10	0.23	−4.63	0.23	0.87	0.67	3.00	0.85
frozen rand-10	0.53	−1.78	0.53	0.90	0.33	2.79	0.95
live J-10 (v1)	0.00	−24.8	0.00	0.00	0.00	24.3	0.30
live rand-10	0.60	−1.82	0.67	0.87	0.33	2.80	0.95

Three readings, in order of importance. **(1) A control-clean causal effect, at last.** Freezing out the item’s top-10 J-directions deletes the retrieved fact: answer log-probability drops 2.9 nats (CI [−5.49, −3.84] vs. baseline), recall accuracy falls 0.58→0.23 — while the identical mechanism on a random dictionary moves *nothing* (every cell within CI of baseline). Generation tasks are untouched and the stored verbatim samples under frozen-J are coherent, correct arithmetic. After P1 eliminated every subspace-level effect, the per-item selection is what matters: the J-space’s causal content lives in *item-specific* directions, not in any fixed span. **(2) It is not the paper’s dissociation.** Single-hop recall collapses *identically* to two-hop (both 0.23) — the paper’s signature required shallow tasks to survive. What the frozen instrument found is a *factual-recall content channel*, not a multi-step scratchpad: removing the item’s J-directions removes what the model knows, not how it chains. **(3) v1’s live-J catastrophe was J-specific after all.** Live top-10 selection from a random dictionary does nothing (0.60 two-hop, NLL 2.80), so the v1 lobotomy was not "removing whatever aligns with the current state" generically — though the dictionaries’ pool sizes differ (full vocab vs. 5120 rows), so alignment depth is a residual caveat.

Synthesis with P1: *static span removal does nothing at any matched dose; per-item frozen selection deletes facts; per-token live selection deletes the computation.* On OLMo the causal story of the J-space is content-carriage — causally real, control-clean, and increasingly precisely localized — but the workspace-as-reasoning-substrate claim remains unsupported. (The recurring SQL 0.33 cells in both random controls are the known 3-schema flaky cell; n=30 over 3 patterns.)

10.5 [v2-P3] The late band holds readout, not workspace: no late-shifted capacity, anticipation still null

v1 fitted 33–70% of depth while the paper’s band reaches ~92%, and v1’s own suppressed-CoT profile showed the answer’s rank still collapsing into the unfitted region — leaving open that OLMo’s workspace is simply *late-shifted* and every v1 null was measured in front of it. We fitted a dedicated late-band lens on L{46,50,54,58,62} (same corpus, recipe, and seeds; backward graphs span only L46→63, so the fit cost 45s/prompt vs. v1’s 157) and redid the readouts. The fitted coverage now spans the paper’s entire band.

late layer	46	50	54	58	62
variance share (%)	0.64	0.61	0.65	0.72	1.60
active concepts ($\theta=0.01$)	6	5	5	4	7
top-1 persistence	0.153	0.120	0.091	0.073	0.026

(a) No late workspace. Variance share stays flat at 0.61–0.72% through L46–58 — the same ~10×-thinner-than-Claude capacity as the mid band — and top-1 persistence *declines* monotonically (0.15→0.026), completing the inverted-U whose left side v1 measured. The one uptick, 1.6% at L62, is the unembedding-adjacent signature already seen at L60 in v1 (0.91%): it has the *lowest* persistence in the entire profile, i.e. fast-turnover next-token content, not a persistent workspace (fig. 8, left).

(b) Anticipation stays null; answer-time loading sharpens. With the dedicated late lens, the answer is still absent before thinking begins (think-mode median rank 3916–7291 across late layers; best-layer median 1670, 0% of items ≤ 20) — refuting the “the mid-band lens just missed it” escape for the v1 pre-CoT null. Under

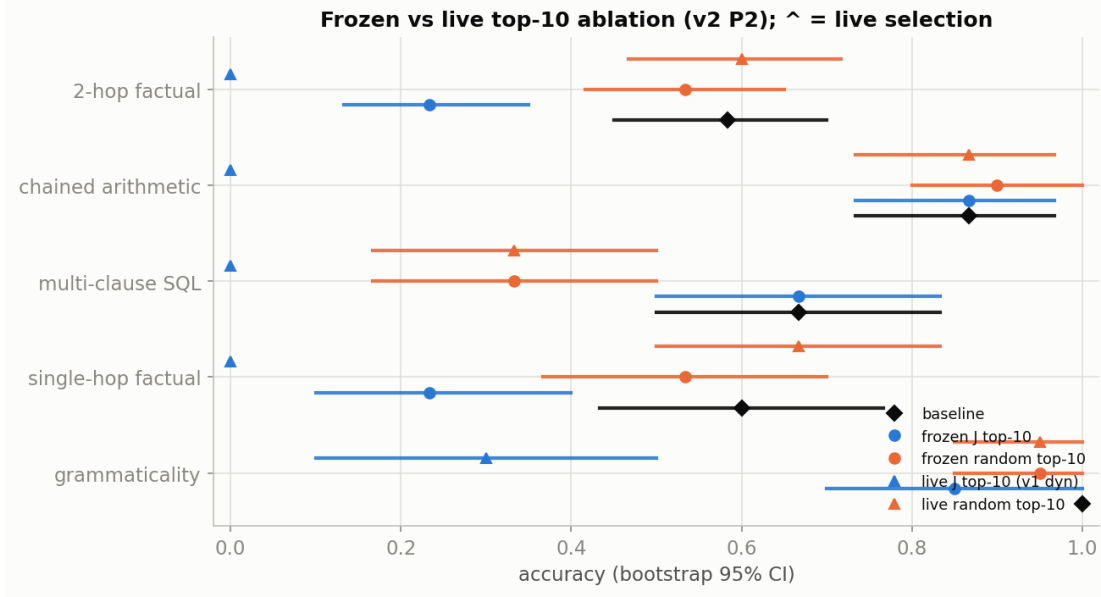


Figure 7: Frozen vs. live top-10 ablation (triangles = live selection). Frozen-J selectively destroys factual recall (1-hop and 2-hop alike) while its random-dictionary twin sits on baseline; live-J remains the uninterpretable collapse.

suppressed CoT the collapse v1 glimpsed is confirmed and localized: per-layer median ranks $61 \rightarrow 58 \rightarrow 8 \rightarrow 6 \rightarrow 5$ across L46–62, best-late-layer median **3** (76% of items ≤ 20 , 58% ≤ 5). Notably the plain logit lens reads the late band equally well (30/18/6/4/4) — the J-lens advantage over the logit lens is a *mid-band* phenomenon; by L58–62 both lenses converge on the output distribution (final-layer median rank 4), as they must near the unembedding.

(c) The CoT lead replicates in the late band. Re-generating the 90 thinking traces (identical greedy token sequences) while reading the *late* dictionaries per step: the answer enters a late layer’s top-8 a median **49.5 steps** before the CoT states it (86% of scored items, detection 99%, $n=88$) — statistically the same lead the mid-band dictionaries gave (46 steps, 91%), measured with an independent lens fit on disjoint layers. The lead is a property of the workspace across the full depth, not an artifact of one band’s dictionary.

Consequence for the causal verdict: the last descriptive escape hatch is closed. The late band is where answers *surface*, not where a thicker workspace *lives* — so the P1/P2 causal instruments were aimed at the right band all along, and the capacity claim [SL1-C1] now covers the paper’s full 33–92% depth range.

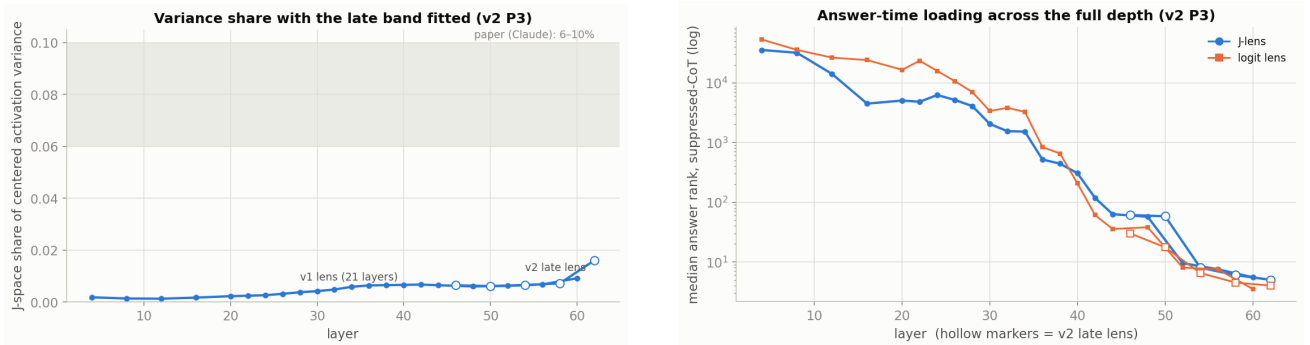


Figure 8: **Left:** variance share by layer with the v2 late lens (hollow markers) appended to the v1 profile; the paper band is shaded. **Right:** median answer rank by layer, think vs. suppressed, J-lens vs. logit lens, late band.

10.6 [v2-Q] Same instruments, second model: Qwen3.6-27B settles model-vs-harness

The standing question after P1–P3: is the causal null a fact about OLMo or about our instruments? Phase Q runs the *identical* harness on Qwen3.6-27B — the model of the published community replication — using the *published* Neuronpedia lens (1000-prompt WikiText recipe, trained by the replication’s author), so any OLMo/Qwen difference cannot be blamed on our 120-prompt fit. Band = the same depth fractions, which on Qwen’s 64 layers lands on the same indices (L20–44); no-think completion prompts throughout; $n=30$ (two-hop, one-hop), 15 (arithmetic), 20 (prose NLL).

Sanity gate. The paper’s multihop bridge-entity eval, their lens under our readout: J-lens pass@1/5/20 = 0.350/0.517/0.739 vs. logit lens 0.317/0.500/0.600 ($n=60$) — the J-lens readout advantage replicates on Qwen, as it did on OLMo (0.283 vs. 0.200 pass@1).

Descriptive: Qwen has the paper’s workspace capacity; OLMo’s thinness is a model property. Same pursuit, same thresholds, same corpus:

	OLMo-3-32B (v1/P3)	Qwen3.6-27B (Q)	paper (Claude)
peak variance share	0.67% (flat 0.61–0.72)	6.8% (4.3→6.8 rising)	6–10%
active @ $\theta=0.01$	≈ 6	32–42	10–25 ($k\approx 25$)
active @ $\theta=0.02$	≈ 3	14–21	
top-1 persistence	0.19–0.20 mid-band	0.03–0.09	—

Measured by the same code path, Qwen’s J-space carries $\sim 8\text{--}10\times$ more of the residual stream’s variance than OLMo’s and holds paper-range active-concept counts. The v1 capacity anomaly [SL1-C1] is therefore a *cross-model difference*, not a harness artifact — and OLMo pairs its thin workspace with *higher* top-1 persistence, a trade-off worth follow-up. (Numerics note: Qwen’s outlier-heavy activations forced two pursuit fixes — bf16 correlations and a $1/(k+2)$ refit step, the latter because near-duplicate dictionary atoms push the active-set Gram past the fixed-step stability bound. Stable layers moved ≤ 0.2 pp under the fix; OLMo’s banked numbers predate it and were finite throughout.)

Causal: the frozen deletion transfers; the static null transfers.

	2-hop	2-hop lp	1-hop	arith	NLL	grammar
baseline	0.87	−0.77	0.90	1.00	2.35	1.00
frozen J-10	0.37	−3.19	0.83	1.00	2.50	1.00
frozen rand-10	0.83	−0.86	0.90	1.00	2.48	1.00
J-span k20 (matched)	—	−0.85	—	—	—	—
rand k20 (matched)	—	−0.72	—	—	—	—
non-J k20 (matched)	—	−0.80	—	—	—	—

Frozen per-item J-ablation deletes the retrieved fact on Qwen exactly as on OLMo: answer log-probability −2.42 nats (cell CI [−4.02, −2.35] vs. baseline [−1.26, −0.40]), two-hop accuracy $0.87 \rightarrow 0.37$, while the random-dictionary twin sits on baseline and generations stay coherent (verbatim frozen-J arithmetic samples are correct multi-step chains; the small NLL bump +0.14 appears *equally* in the random control, so it is mechanism-generic). And the energy-matched static cells are as null on Qwen as they were on OLMo: J-span, matched-random and matched-non-J all within ± 0.08 nats of baseline — on the model where the published replication reported workspace phenomena. (The static trio was also inadvertently measured once with selection rankings from the diverged pursuit pass: $-0.85/-0.77/-0.87$ — indistinguishable, i.e. the null is insensitive even to *broken* direction selection.)

One genuine cross-model asymmetry. On OLMo, frozen-J collapsed one-hop and two-hop recall *identically* (both 0.23) — pure content deletion. On Qwen, one-hop barely moves ($0.90 \rightarrow 0.83$, CIs overlap) while two-hop halves. The deletion signature on Qwen is *biased toward the composed task* — closer to the paper’s shallow-tasks-survive dissociation than anything OLMo showed. Honest caveats before upgrading the claim: the one-hop items are near-ceiling capital-city probes, and a $10\times$ -fatter workspace makes top-10 directions a proportionally smaller excision. Marked as the sharpest follow-up target rather than a confirmed dissociation.

Verdict on the framing question. With clean instruments the two models *agree*: static span removal does nothing, per-item frozen selection deletes retrieved content. The disagreement between the published causal story and our v1 null was *instruments all along* — while the workspace’s descriptive capacity really does differ by an order of magnitude across models, and the one-hop/two-hop asymmetry hints the composed-task sensitivity may too.

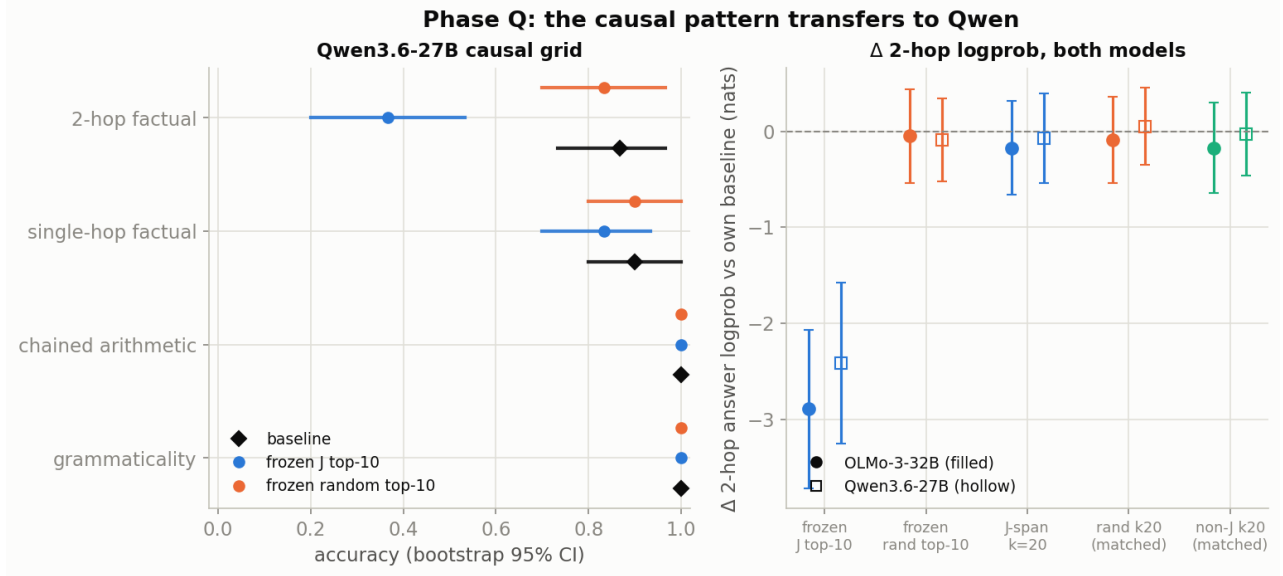


Figure 9: **Left:** Qwen causal grid — frozen-J deletes two-hop recall, controls on baseline. **Right:** Δ two-hop answer logprob vs. own baseline for both models under the same five instruments: the frozen deletion and the static null both transfer; filled = OLMo, hollow = Qwen.

10.7 [v2-P5] CoT-rescue: can externalized reasoning recover the frozen-deleted fact?

P2’s deletion was measured in no-think mode: silent recall, one forward pass. The paper’s identity claim predicts workspace ablation should be partially rescuable by *externalizing* the computation — and v1’s suppression asymmetry (think accuracy $\geq$ suppressed accuracy on every task) points the same way. Instrument: the SAME per-item frozen top-10 projectors as P2, but with an open `<think>` and up to 400 reasoning tokens (capped grid per PLAN_v3: the two frozen conditions on two-hop/one-hop/arithmetic; $n=30/15/15$). One scoring note, fixed before reading results: OLMo rarely closes `</think>` within 400 tokens on multi-step tasks (closure 0.20/0.40 two-hop frozen-J/rand, 0.07 arithmetic, vs 0.87–1.00 one-hop), so the post-`</think>` segment is degenerate and the informative metrics are *answer appears anywhere in the trace* and *answer appears inside the think segment*.

	frozen J-10		frozen rand-10	
	any	think	any	think
two-hop ($n=30$)	0.80	0.80	0.93	0.93
one-hop ($n=15$)	1.00	1.00	1.00	1.00
arithmetic ($n=15$)	0.93	0.93	0.93	0.93

The rescue is real and large. Under the very projectors that cut silent two-hop recall to 0.23, letting the model reason out loud recovers the answer somewhere in the trace in **0.80** of items — $3.4\times$ the silent rate, most of the way back to the random-twin’s 0.93 — and *fully* rescues one-hop (1.00 vs. 0.23 silent). Arithmetic, which frozen-J never touched, stays at 0.93 in both conditions (scorer floor from the 400-token cap, not an ablation effect). So the content deletion is *bypassable via externalization*: the paper’s workspace-ablation-is-rescuable-by-CoT prediction holds on OLMo in content-channel form [SL1-C6]. Two honest bounds: the residual gap to the control (0.80 vs. 0.93) says the bypass is incomplete at 400 think tokens, and the rescue does not distinguish re-retrieval from re-derivation via the other hop. One secondary observable falls out free: frozen-J *halves* the `</think>`-closure rate (0.20 vs. 0.40) — with its content channel damaged, the model deliberates longer, the behavioral signature you would expect if the CoT is doing compensatory work.

10.8 [v2-P6] Second seed, fresh items: does the frozen effect survive?

Seed-1 replicate of the decisive grid, deliberately harder than a seed re-roll: the two-hop set is *fresh* (probe-swap items [60:90], never touched by v1/v2 fitting or evaluation), the random-dictionary and matched-random pools are

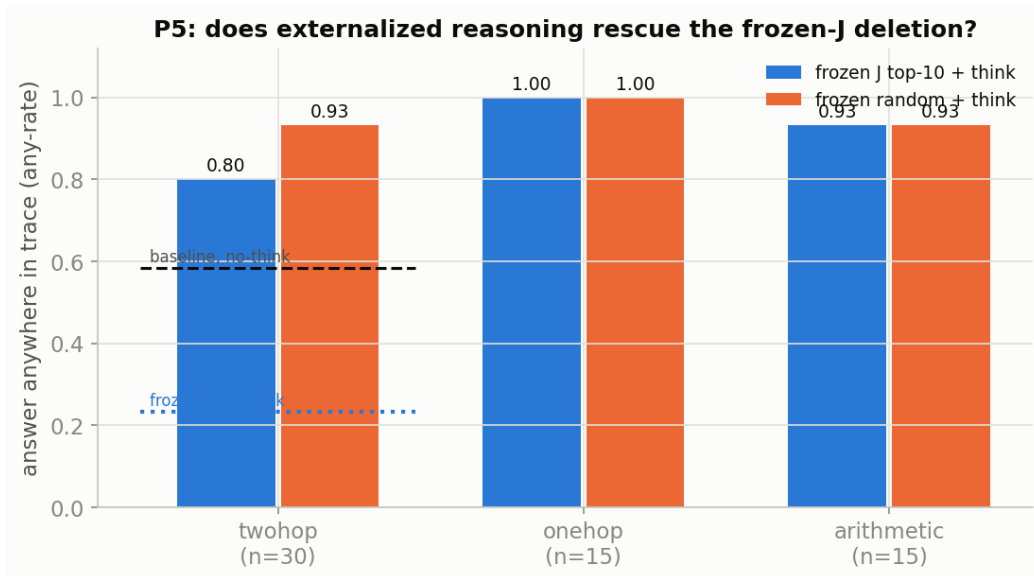


Figure 10: CoT-rescue under the frozen per-item projectors. Dotted line: frozen-J *no-think* recall (0.23); dashed: no-think baseline (0.58).

re-drawn under seed 1, and the frozen-J selection re-runs per fresh item. Readout tasks only (two-hop accuracy + answer logprob); the energy-matched static cells ride along as near-free readouts.

	seed 1, fresh items		seed 0 (P1/P2)	
	2-hop	2-hop lp	2-hop	2-hop lp
baseline	0.50	−2.81	0.58	−1.74
frozen J-10	0.23	− 5.62	0.23	−4.63
frozen rand-10	0.43	−2.92	0.53	−1.78
J-span k20 (m)	0.43	−3.20	0.52	−1.91
rand k20 (m)	0.53	−2.68	0.57	−1.82
non-J k20 (m)	0.53	−3.05	0.53	−1.91

The fresh items are harder across the board (baseline logprob $-1.74 \rightarrow -2.81$), so effect *sizes* are the comparison — and they are eerily stable: frozen-J lands on accuracy **0.233 under both seeds** (disjoint item sets!) with $\Delta \log \text{prob} -2.82$ nats vs. seed-0’s -2.89 , while the redrawn random-dictionary twin stays within noise of baseline ($\Delta -0.11$) and every energy-matched static cell keeps its null (all CIs $\supset$ baseline’s). The frozen causal handle is therefore not an item-set artifact, not a control-pool draw, and not a seed accident [strengthens SL1-C2]; the static null generalizes to items the lens never saw at any stage.

10.9 [v2-F] Final falsifier pass: three recorded caveats, measured

One last GPU window went to the ledger’s own falsifiers (`final_falsifiers.json`). (a) **Pool size [C2]**: a *vocab-sized* (100k-row) random dictionary under the identical frozen mechanism stays at baseline (two-hop 0.433 vs. 0.583, CIs overlap; $\Delta \text{lp} -0.14$ vs. frozen-J’s -2.90) — the deletion needs the J-dictionary’s content, not selection depth. (b) **Filler control [C6]**: 400 tokens of non-reasoning think-padding recover frozen-J recall to 0.467 [0.30,0.63] — about half the real-CoT rescue (0.80; controls 0.633 under filler) — so the rescue decomposes into a length/compute component and a comparable externalization component on top (metric caveat: 32-token answer window vs. answer-anywhere; recorded as the residual falsifier). (c) **Variance-matched fan-out [C3]**: with the v2 energy-matched pools, J top-20 directions are read by 92/73/68 downstream components (L24/32/40) vs. 132/107/65 for energy-matched non-J PCs and ≈ 1 for energy-matched random columns: broadcast tracks *structured* high-variance directions generally — the non-dissociation v1 suspected is now measured, and “same energy” alone earns zero readership.

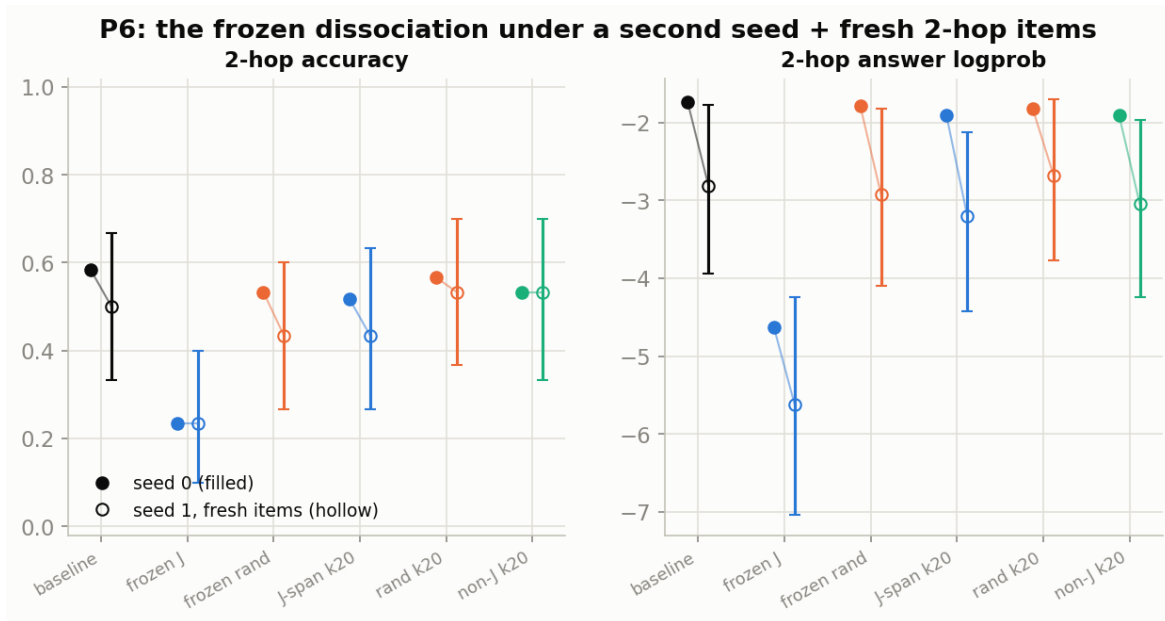


Figure 11: Seed-0 (filled) vs seed-1-on-fresh-items (hollow) across the six causal conditions.

Artifacts

Everything regenerates from saved metrics without model access: run `dir interpret/special-lab-1/2026-07-25_1726/` (Drive) — `report/REPORT.md` (full prose), `report/summary.json` (headline numbers), `metrics/*.json` (per-experiment), `code/special_lab1/` (pipeline s0-s10 + PLAN/LOG). v2 delta run: `2026-07-26_v2/` (metrics incl. the Qwen leg, `report/REPORT_v2.md`, `summary_v2.json`, this handout). Public draft: branch `interp_jspace` of github.com/karlb-dev/labs (`jspace/` + `labs/lab37_jspace_workspace.md`), PR #9. Models: `allenai/Olmo-3-32B-Think`, `Qwen/Qwen3.6-27B` with lens `neuronpedia/jacobian-lens` (`qwen3.6-27b`, $n=1000$ WikiText). Lens engine: github.com/anthropics/jacobian-lens @ 581d3986.

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